

GENERAL SPECIFICATION G4.64

Concentric Tube Heat Exchangers for Sludge

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G4.64 CONCENTRIC TUBE HEAT EXCHANGERS FOR SLUDGE

G4.64.1 General

- (a) This Specification defines mandatory requirements for the design, materials, fabrication, assembly, delivery, installation, testing and commission of concentric tube heat exchangers.
- (b) Concentric tube heat exchangers complete with all appurtenances and accessories shall be the end product of one responsible manufacturer or responsible system supplier.
- (c) Unless otherwise indicated, the Contractor shall obtain each heat exchanger system from the responsible supplier of the equipment.
- (d) The Contractor shall supply and co-ordinate all components and accessories as necessary to place the equipment in operation in conformance with the specified performance, features and functions indicated without altering or modifying the Contractor's responsibilities under the Contract Documents.
- (e) The Contractor is responsible to the Board for providing the equipment systems as specified herein.

G4.64.2 Design Codes and Standards

Applicable design codes and standards shall be as follows:

- (a) ASME Code, Section II (2017) - Material Part D Properties (Metric).
- (b) ASME Code, Section V (2017) - Non-destructive Examination.
- (c) ASME Code, Section VIII - Division 1 (2017), Rules for Construction of Pressure Vessels.
- (d) ASME Code, Section IX (2017) - Qualification Standard for Welding, Brazing, and Fusing Procedures; Welders; Brazers; and Welding, Brazing, and Fusing Operators.
- (e) ASME B31.3 (2016) - Process Piping.
- (f) ASCE, Minimum design loads for building and other structures.
- (g) ASME B16.47 (2017) - Large Diameter Steel Flanges: NPS 26 through NPS 60 Metric/Inch Standard.
- (h) AWS - American Welding Society.

G4.64.3 Submittals

- (a) As a minimum, the following shall be submitted:
 - (i) Make, model, and weight of each heat exchanger equipment assembly.
 - (ii) Complete catalogue information, descriptive literature, specifications, technical datasheets and identification of materials of construction.
 - (iii) Detailed structural and mechanical drawings showing the equipment dimensions, size, and locations of connections and weights of associated equipment.

- (iv) Factory and Field Painting Systems Proposed: Include manufacturer's descriptive technical specifications and data sheets.
- (v) Factory Functional Test Report.
- (vi) Field Performance Test Report.
- (vii) Site Test Procedure including comprehensive listing of test apparatus and appurtenances
- (viii) Certificate of Compliance with ASME Code Section VIII, Division 1 (2017).
- (ix) Manufacturer's Certification of Compliance that the factory finish system is identical to the requirements specified herein.
- (x) Special shipping, storage and protection, and handling instructions.
- (xi) Manufacturer's installation instructions.
- (xii) Routine maintenance requirements prior to plant start-up.
- (xiii) Manufacturer's Certificate of Proper Installation
- (xiv) Operation and Maintenance Manuals
- (xv) Manufacturer's recommended spare parts list to maintain the equipment in service for a period of 1 year and 5 years. Include a list of special tools required for checking, testing, parts replacement, and maintenance with current price information.
- (xvi) Listing of special tools, materials, and supplies provided with equipment for use prior to and during start-up and for future maintenance.
- (b) Refer to General Specification G1.2 Information Management for further requirements on submittals.

G4.64.4 Spare Parts and Special Tools

- (a) Spare parts and special tools requirement shall be as per the tender Bill of Quantities.

G4.64.5 Concentric Tube Heat Exchangers Requirement

- (a) Heat exchanger shall be of a standard and industry proven design, sized for the full range of operating duties, temperatures and conditions specified in the Particular Specifications,
- (b) Selection of heat exchanger shall comply with the minimum design life of 25 years.
- (c) The following types of heat exchanger shall be used:
 - (i) Concentric tube type, designed with no sharp bends or obstructions in internal passages. Unit shall provide full counter current heat transfer. With full bore flanged removable bends at the end of each heat exchanger length for cleaning.

- (ii) Heat exchanger cleaning shall be by means of high pressure water-jet at 700 to 1200 barg. Depending on the nature and severity of the fouling, hot water with or without chemical agents may be used. In addition, rigid or flexible jetting lances may be employed to direct the cleaning jet into the heat exchanger tubes. The heat exchanger supplier/manufacturer recommendations for cleaning the specific model of heat exchangers offered shall be submitted for review by the S.O.. This shall include full details of water flowrate, pressure, temperature, cleaning equipment, appurtenances and cleaning agents, etc..
- (d) Couplings for temperature and pressure gauges, backflush connections on sludge side, and 50 mm drain with valves shall be provided.
- (e) Pressure indicators and transmitters, temperature indicators and transmitters, and any other instruments necessary for full functionality and safe operation of the heat exchangers, shall be provided. All instruments shall be provided complete with isolation valves or similar to allow for maintenance.
- (f) Pressure relief valves shall be provided.
- (g) Handhole cleanout ports and inspection covers, where applicable, shall be provided.
- (h) Flanges for all heat exchanger connections and adjoining fittings shall comply with BS EN 1092 and shall be rated and drilled in accordance with the flange pressure ratings of the connected process piping.
- (i) Austenitic stainless steel U-bends shall be solution annealed after bending.
- (j) Heat exchangers shall be post weld heat treated when specified on the data sheet(s) or when required by the Code. When post weld heat treatment is required, the whole exchanger shall be so treated at one heating. If a single heat treatment is impossible, the Contractor shall so state this exception in his proposal and describe his proposed post weld heat treating procedure.
- (k) Welding of galvanised steel to any austenitic stainless steel heat exchanger is prohibited. Welding of painted carbon steel having zinc rich primers to austenitic stainless steel heat exchangers is prohibited, unless a cleaning procedure is used that is approved by the S.O. prior to welding.
- (l) Strength welds shall be in accordance with ASME Section VIII, Division 1 (2017), Appendix A, Fig. A-2 Type 3 or 4. The "a" dimension must equal or exceed 1.4 (t). The ends of the tubes shall be free of deposited weld metal.
- (m) Welded tube-to-tubesheet joints shall be preheated to 93°C minimum when the tubesheet is P-1 material greater than 1-1/4 inches thick.
- (n) The Brinell Hardness of the weld metal and heat affected zone (HAZ) of pressure retaining welds shall be made in accordance with API 660. Hardness requirements for H₂S/Amine services shall be specified separately.
- (o) As a minimum control, spot radiographic inspection in accordance with the ASME Code is mandatory for all heat exchanger butt welds and 100 percent radiographic inspection shall be performed when specified in the data sheets.

- (p) All weld bevels in material 1-½ inch thick and over shall be examined by the magnetic particle or liquid penetrant method. Any laminations exceeding the limits specified in Paragraph 9.3 of ASME SA-20 shall be repaired.
- (q) All bolted connections such as channels, covers, blinded flanges, etc., shall be torqued to the proper values before shipment. When design temperatures are over 340°C, an approved high temperature thread lubricant shall be used.

G4.64.6 Heat Exchanger Performance

- (a) The heat exchangers shall be sized for the maximum duties as defined in the heat and mass balances required to be submitted.
- (b) They shall be configured to enable control of the temperature of the sludge streams and subsequently the bulk temperature of the digesters.
- (c) An installed standby heat exchanger shall be provided to take over the duty role of any one of the duty heat exchangers.
- (d) The associated flushing system of each heat exchanger unit shall be clearly depicted in the manufacturer's P&IDs/schematic drawings and general arrangement drawings. The flushing system shall prevent solidification of residual sludge in the heat exchanger that is not in use.
- (e) Where applicable, dual-stage heat exchanger of the series configuration may be employed but such that the performance and coefficient of heat transfer shall not be inferior to single stage heat exchangers.
- (f) Design pressure of each heat exchanger shall be the maximum operating pressure added with 10 percent or 1.72 Barg, whichever is greater.
- (g) Where not specified in datasheets, design pressure of heat exchangers shall not be less 3.45 Barg.
- (h) Unless otherwise specified, the minimum tube side water flow velocity shall be 1.22 m/s for water-cooled heat exchangers.
- (i) The total pressure drop shall include the pressure drop at the inlet nozzle, outlet nozzle and bundle entry and exit losses.
- (j) Space provision shall be included to add on heat exchanger units or modify the heat exchangers where necessary with respect to heat and mass balances.
- (k) The Contractor shall consolidate information of cleaning agents recommended by the heat exchanger manufacturer for cleaning of fouled heat exchangers and removal of scale. These cleaning agents shall not cause the dissolution nor affect the chemistry of the heat exchanger parent material.

G4.64.7 Materials of Construction

- (a) Sludge side: SS316L
- (b) Water side: SS316L
- (c) Tie-rods (if required): SS316L
- (d) Tube support baffles (if required): SS316L
- (e) Tube sheet/gasket: Graphite/PTFE/Mica facing, Graphite/PTFE/Mica filled, with SS304/SS316 winding

- (f) Expansion joint (if required): SS316L

G4.64.8 Accessories

- (a) Lifting Lugs: For equipment entity weighing over 25 kg.
- (b) Anchor Bolts: To be minimally 12 mm diameter of stainless steel SS316 material and sized by equipment manufacturer

G4.64.9 Factory Finishing

- (a) Surface preparation, priming, intermediate and finish coats for protection of heat exchanges and appurtenances shall be according to General Specification G1.5 Painting and Protective Coatings.
- (b) The Contractor shall ensure that any special painting or coating system applicable shall not negatively affect the heat transfer characteristics and efficiency of the heat exchanger.
- (c) Coating applied to the interior of heat exchanger tubes shall exhibit characteristics of minimal resistance to flow, protect the coated surfaces from corrosion and not add impurity to the fluid.

G4.64.10 Source Quality Control

- (a) The Contractor shall ensure that all heat exchanger units shall be inspected for proper installation, alignment, connection, leakage and quiet operation by the supplier/manufacturer.
- (b) After hydrostatic test, the equipment shall be completely drained and compressed air dried for approximately 30 minutes so that no liquid remains in the shell or tubes. The Contractor shall ensure that the supplier take extra precautions to ensure draining is complete and surfaces are dry for heat exchangers fabricated of martensitic, ferritic and austenitic stainless steels (AISI 300 and 400 series). Flange faces and other exposed machine carbon steel surfaces must be coated with a suitable rust preventative.
- (c) Prior to acceptance of each heat exchanger installation, following the verification and witnessing by the supplier/manufacturer, each unit shall be operated to demonstrate its ability to operate without overloading, or excessive vibration.
- (d) Performance tests shall be carried out for each heat exchanger unit as follows:
 - (i) Test for a continuous 30-minute period on sludge and cooling water channels without malfunction or leakage.
 - (ii) Hydrostatic test pressure of heat exchangers shall be 1.3 times the maximum allowable working pressure multiplied by the lowest stress ratio (LSR) for the pressure-boundary materials of which the vessel is constructed and any other applicable requirements of sub-section UG-99 of ASME Boiler and Pressure Vessel Code Section VIII Div. 1.

G4.64.11 Manufacturer's Services

- (a) Comply to General Specification G2.25 Manufacturers' Services.

END OF SECTION